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7th Grade STEM is designed to develop scientifically literate, inquisitive students with a solid understanding of basic life science concepts and inquiry skills. It is focused on ways to unite our world view as citizens of the planet, nurture our understanding of how the biological systems are interrelated and inspire the next generation of scientists and engineers.

The year begins with a short review of the scientific method and the engineering design process as well as the principal tools and techniques used by scientists and engineers. Then we launch into a study of how we perceive the world. We will investigate how our senses interface with our brain, how the brain works, grows, and learns. Each student will conduct a dissection of an organ to gain first-hand knowledge of dissection skills. After Winter Break, our students will learn about the form and functions of cells, focusing on the role of bacteria and viruses in our world. This unit culminates in a formal debate about whether Viruses are alive. Following Spring Break, the year continues with a study of human body systems, heredity, and genetics and ends with an examination of human reproduction and life in the womb.

2025 – 2026 Scope and Sequence:

August-June

All unit dates are *projected* and may be subject to change.

Unit 1: Why STEM?

5 Aug – 30 Aug

Key Concept: Systems

Related Concept(s): Patterns, Form, Function

Global Context: Science and Technical Innovation

Inquiry statement: The Scientific Method and the Engineering Design Process allow us to investigate the form and function of patterns observed in natural systems or designs.

Inquiry questions:

Factual: How do scientists and engineers use the Scientific Method and the Engineering Design Process?

Conceptual: How does experimental science and engineering expand our body of knowledge?

Debatable: Can the Scientific Method or the Engineering Design Process be used to answer any question or solve any problem?

Content

- **Know:** Scientific Method, Engineering Design Process, Human Water Cycle
- **Understand:** How the Scientific Method is used to answer questions. The linkage between the natural water cycle and the human water cycle.
- **Do:** Lab Safety, Scientific Method/Engineering Design Process, variables and hypothesis formulation, Lab report format, Water Quality Lab.

Summative Assessment Tasks

Hawaiian Water Cycle Poster (Crit. A/D)

Unit 2: By the Dawn's Early Light

2 Sep – 18 Dec

Key Concept: Perspective

Related Concept: Evidence and Patterns

Global Context: Identities and Relationships

Statement of inquiry: A personal perspective is formed from learning based on evidence, facts, observations, and ideas.

Inquiry questions:

Factual: How does your brain make sense of the world?

Conceptual: How does evidence shift perspective?

Debatable: How does your perspective relate to your values (perception of right or wrong)?

Content

- Know: Anatomy and physiology of the human brain and eye,
- Understand: The uniqueness of each of our brains, responses by the brain differ between individuals, learning is an important brain function, and there are factors that affect learning, the ability of the brain to learn is not fixed.
- Do: Dissection skills, designing, conducting, and communicating the results of a scientific investigation, collecting, interpreting, and analyzing data.
- Labs: Mass, Volume, Density, Learning/Memory Brain Investigation

Summative Assessment Task(s)

- Technology and Perception Essay (Crit A,D)
- Brain Memory/Learning Inquiry (Crit B,C)

Unit 3: An Invisible Army.

6 Jan – 13 Mar

Key concept: Relationships

Related concept: Function and Models

Global Context: Scientific and Technical Innovation

Statement of inquiry: Examining the relationships between organisms allows us to model their functions to better understand how they interact.

Inquiry questions:

Factual: What are the differences between plant, animal, and bacterial cells?

Conceptual: How do bacteria affect humans and the environment?

Debatable: Who is most likely to survive: humanity or bacteria?

Content

- Know: Cell organelles and functions, the differences among plant, animal, and bacterial cells, the differences between bacteria and viruses
- Understand: The process of osmosis, bacteria are both beneficial and harmful
- Do: Microscope skills, designing, conducting, and communicating the results of a scientific investigation, research skills
- Labs: Respiration/Combustion Lab, Osmosis Investigation

Summative Assessment Task(s)

- Cell Analogy Poster (Crit A,D)
- Osmosis Lab (Crit B,C)
- Virus Debate (Crit A,D)

Unit 4: Body Systems, Genetics and Reproduction.

30 Mar – 3 Jun

Key Concept: Change

Related Concept(s): Interaction and Transformation

Global Context: Orientation in Space and Time

Inquiry statement: Observing changes in interactions allows scientists to detect and understand the transformations in living organisms.

Inquiry questions:

Factual: What is DNA and how does it work?

Conceptual: How does genetics relate to evolution?

Debatable: Does evolution discount the existence or workings of God?

Content

- **Know:** Human Body Systems, Mendel's Laws of Heredity, DNA/RNA transcription, Protein synthesis, evidence of evolution.
- **Understand:** How the human body systems function to support each other. How genetics is the underpinning of the evolution of life on Earth.
- **Do:** Body Systems, Blood Lab, Rice Babies, Punnet Squares, Sesame Street Genetics

Summative Assessment Tasks

Rice Babies (Crit A,D)

Blood Lab (Crit. B,C)

Sesame Street Genetics (A,D)

STEM Assessment Criteria

	Achievement Levels
A. Knowing and Understanding	0-8
B. Inquiring and Design (1 st half of Experimental/Design Cycle)	0-8
C. Processing and Evaluation (2 nd half of Experimental/Design Cycle)	0-8
D. Reflecting on the Impacts of Science/Engineering	0-8

Student activities will be assessed against the criteria listed above. End of quarter letter grades will be assigned based on student learning progress over the quarter.

Contact Information: email deachus@starofthesea.org

- Parents with questions or concerns may directly contact any instructor.
- Students are encouraged to use study hall or lunch recess to obtain additional learning support. In addition, **the Innovations Lab will be open each Wednesday afternoon (except Mini-days) until 4 pm for any student needing help or to finish lab work.**
- Students and/or parents should use email to contact their instructor to make arrangements for additional learning opportunities.